

Experiment: K-Means and Hierarchical Clustering

Dataset Source

- Dataset Used: Mall Customer Segmentation Dataset

<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>
- Direct CSV (used in implementation):

https://raw.githubusercontent.com/sharmaroshan/Mall-Customer-Segmentation-Dataset/master/Mall_Customers.csv

Dataset Description

The dataset contains information about mall customers and is used for customer segmentation.

- **Total Records:** 200
- **Number of Features:** 5
- **Target Variable:** None (unsupervised learning)
- ♦ **Features:**
 - CustomerID – Unique identifier
 - Gender – Male/Female
 - Age – Customer age
 - Annual Income (k\$) – Income in thousand dollars
 - Spending Score (1–100) – Customer spending behavior
- ♦ **Characteristics:**
 - Mix of categorical and numerical data
 - No predefined labels
 - Suitable for clustering algorithms

- Helps in identifying customer groups

3 Mathematical Formulation of the Algorithm

◆ K-Means Clustering

K-Means aims to minimize the distance between data points and their cluster centroids.

$$J = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

Where:

- C_i = i th cluster
- μ_i = centroid of cluster

◆ Hierarchical Clustering

Hierarchical clustering builds a tree (dendrogram) using distance metrics.

Distance formula (Euclidean):

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

Linkage methods:

- Single linkage
- Complete linkage
- Ward linkage

4 Algorithm Limitations

◆ K-Means:

- Requires predefined number of clusters (K)
- Sensitive to outliers
- Works poorly with non-spherical clusters
- Depends on initial centroid selection

◆ Hierarchical Clustering:

- Computationally expensive for large datasets
- Difficult to determine optimal clusters
- No re-adjustment once clustering is done

5 Methodology / Workflow

◆ Steps:

1. Load dataset
2. Preprocess data (encode categorical values)
3. Select relevant features (Income & Spending Score)
4. Apply K-Means clustering
5. Use Elbow Method to determine optimal K
6. Apply Hierarchical Clustering
7. Plot dendrogram
8. Visualize clusters
9. Analyze results

◆ Workflow Diagram:

Dataset → Preprocessing → Feature Selection →
K-Means → Elbow Method → Cluster Formation
↓
Hierarchical Clustering → Dendrogram → Cluster Analysis

6 Performance Analysis

Since clustering is an unsupervised learning technique, accuracy and confusion matrix cannot be used.

◆ Metrics Used:

- **WCSS (Within Cluster Sum of Squares):**
Measures cluster compactness (lower is better)
- **Silhouette Score (~0.5–0.6):**
Indicates quality of clustering
 - Close to 1 → good clustering
 - Around 0.5 → moderate but acceptable

◆ Observations:

- Optimal number of clusters: **K = 5** (from Elbow Method)
- Clusters formed are meaningful and well-separated
- Customer groups identified based on income & spending

◆ Interpretation:

Customers are grouped into segments such as:

- High income – high spending
- High income – low spending
- Low income – high spending
- Low income – low spending

7 Hyperparameter Tuning

◆ K-Means:

- Parameter tuned: **Number of clusters (K)**
- Method used: **Elbow Method**

- Optimal value: **K = 5**

◆ Hierarchical Clustering:

- Parameter tuned: **Linkage method**
- Best method: **Ward linkage**

◆ Impact:

- Improved cluster separation
- Better visualization and interpretation
- Reduced intra-cluster distance

Conclusion

Both K-Means and Hierarchical Clustering successfully grouped the dataset into meaningful clusters. K-Means provided efficient and fast clustering, while hierarchical clustering offered a clear visual representation through dendrograms. These clustering techniques are highly useful for customer segmentation and business decision-making.